

ASSIGNMENT – 1

C Programming – Unit 1

B.Tech. 1st Year (SCSE) – IILM University

Instructions

- Answer all 20 questions.
- Write answers clearly and use suitable examples, diagrams, or flowcharts wherever required.
- For programming questions, write syntactically correct C programs and explain the important steps.
- Keep answers concise but complete.

CO1: Apply fundamental programming constructs such as loops, conditional statements, arithmetic and logical operations to develop simple algorithms for solving computational problems.

Part A – Short Answer Questions

Answer briefly. Focus on definitions, basic concepts, and important terminology.

1. What is a computer? Mention any four important characteristics of a computer.
2. What is memory hierarchy? List its major levels in order from fastest to slowest.
3. What is an Operating System? Mention any two important functions of an Operating System.
4. Differentiate between a compiler and an interpreter.
5. What is an algorithm? State any two characteristics of a good algorithm.
6. What are tokens in C? Name the major types of C tokens.
7. Differentiate between keywords and identifiers in C with one example of each.
8. What is type conversion in C? Differentiate between implicit and explicit type conversion.

Part B – Medium Answer Questions

Answer with suitable explanations. Use comparisons, examples, or diagrams wherever helpful.

9. Explain the classification of computers based on size and capability. Give suitable examples.
10. Draw and explain the memory hierarchy. Compare the levels in terms of speed, capacity, and cost.
11. Explain the basic operational cycle of a CPU while executing an instruction.
12. Explain the major stages of program development, from problem definition to testing and debugging.
13. Explain step-form, pseudocode, and flowchart as different ways of representing an algorithm. Give a simple example.
14. What is structured programming? Explain how problem decomposition helps in developing a program.

Part C – Long Answer / Application Questions

Answer in detail. Include suitable C programs, step-by-step evaluation, or diagrams wherever required.

15. Explain the structure of a C program with a suitable example. Identify and explain the purpose of header files, main(), declarations, statements, and comments.
16. Explain C data types, variables, and constants with suitable examples. Also differentiate between variables and constants.
17. Explain arithmetic, relational, logical, assignment, increment/decrement, conditional (ternary), and bitwise operators in C with simple examples.
18. Explain operator precedence and associativity in C. Demonstrate their effect by evaluating suitable expressions step-by-step.
19. Write a C program to input two numbers and display their sum, difference, product, quotient, and remainder. Explain the operators used.
20. Write a C program that demonstrates implicit and explicit type conversion. Explain the output and why the conversions occur.

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